



Nunavik Nickel Project
Water Stewardship Annual Report - 2023



INTRODUCTION

Canadian Royalties Inc (CRI) mines nickel and copper ores at its Nunavik Nickel Project (NNiP) mining complex. This mining project is located approximately 80 km west of Kangiqsujaq and about 140 km southeast of Salluit in Northern Quebec. PNNi began in 2008 with the discovery of the Mesamax, Expo, Méquillon and Ivakkak deposits. The Allammaq and Puimajuq deposits were subsequently added to the Global Certificate of Authorization No. 3215-14-007 issued by the Quebec government.

An Environmental and Social Impact Assessment (ESIA) was carried out in 2007 to characterize the receiving environment, including the water environment, and assess the potential impacts of PNNi on it. An environmental monitoring program (EMP) and a set of mitigation measures were developed and have been implemented for several years.

As a member of the Quebec Mining Association, CRI adheres to the Mining Association of Canada's *Towards Sustainable Mining* (TSM) initiative. CRI implements the Water Stewardship Protocol by integrating it into its activities and the PNNi Environmental Management Program. CRI is also committed to ensuring that its actions are consistent with the protocol's strategic framework, presented in the appendix to this report.

The Environment Department and the Water and Residuals Management Department have defined roles and responsibilities for operational water management and watershed planning. The purpose of this document is to report on the status of CRI's performance against established water management objectives and targets for the year 2023.

A handwritten signature in blue ink, reading "Stéphane Twigg".

Stéphane Twigg, Environmental Director

RESULTS 2023

Target 1: Water usage rate per tonne of ore processed

The water that feeds NNiP's ore processing plant to transform the ore into copper and nickel concentrates comes from Bombardier Lake. CRI has set a utilization rate target of 0.34 m³ of water per tonne of ore processed, based on maximum recirculation of mine water with respect to technical-economic feasibility.

Total water withdrawal from Bombardier Lake is included in NNiP's EMP (Monitoring # 1); results are reported to the MELCCFP and Inuit communities in the annual EMP report. CRI pays annually the royalties required under the *Règlement sur la déclaration des prélèvements d'eau* to the Quebec government

Table 1: Performance for target water use rate per tonne of ore processed

Month	Target (m ³ /t ore)	Result (m ³ /t ore)
January	0,34	0,63
February	0,34	0,55
March	0,34	0,58
April	0,34	0,53
May	0,34	0,53
June	0,34	0,38
July	0,34	0,32
August	0,34	0,47
September	0,34	0,40
October	0,34	0,35
November	0,34	0,33
December	0,34	0,41

The target was not met for most months of 2023, due to the delay in installing a water softening recirculation treatment unit, linked to the process of obtaining ministerial authorization. As a result, the quantity of recirculated water was lower than planned, leading to a higher utilization rate of fresh water. The total water withdrawal for 2023 is 807,360 m³. Installation of the softening unit is scheduled for autumn 2024.

Target 2: Quantity of water treated by mining water treatment units (WTP)

For each of the NNiP mining sites, runoff water is collected by one or more collection basins and treated by a WTP before discharge into the environment. PNNi has four WTPs; the Mesamax WTP treats water from two other mine sites (Allammaq and Puimajuq). Based on the volumes of water to be treated annually at each mine site, specific targets have been set for each one.

Table 2: Performance in terms of target volume of water treated for each WTP

WTP	Target (m) ³	Earnings (m) ³
Expo	1 300 000	1 322 072
Mesamax	300 000	357 008
Méquillon	200 000	267 372
Ivakkak	80 000	90 840

Every one of our targets has been met or even surpassed.

Target 3: Compliance of mine effluent discharges

The quality of the water treated and then discharged (effluent) into the environment by NNiP's mining water treatment units (WTP) is included in the EMP (Monitoring # 3). This quality monitoring is carried out at different frequencies:

- Continuous pH and flow measurement
- Analyses in an accredited laboratory 3 times/week, 1 time/week and 1 time/month for various groups of physic-chemical parameters (metals, suspended solids, nitrates, sulfates, etc.).
- Monthly bioassays for acute lethality on rainbow trout and *Daphnia Magna* in an accredited laboratory.

CRI's target is 0 non-compliance with all requirements applicable to each WTP (Metal and Diamond Mining Effluent Regulations, Directive 019, Depollution attestation, WTP operating permits).

Table 3: Efficiency of compliance with applicable requirements for each WTP

WTP	Target	Results
Expo	0	1
Mesamax	0	0
Méquillon	0	0
Ivakkak	0	0

The Expo final effluent showed 1 non-conformity in October 2023. The results and corrective plan have been communicated to government authorities.

OBJECTIVES 2024

In addition to maintaining the 2023 targets, the main water management objectives for 2024 are:

- 1) Installation of a water softening unit to increase water reuse in the ore concentration process.
- 2) Finalization of design documents for a new sanitary water treatment plant at the Expo camp. The new plant's more advanced technology will enable discharge of an even more purified effluent.
- 3) Pilot testing of the use of water from mine site collection basins for underground drilling as a replacement for fresh water from Bombardier Lake, with the aim of achieving 100% reuse of mine water during the summer months for this type of water consumption.

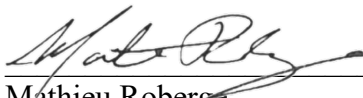
APPENDIX: Water stewardship Policy of Canadian Royalties inc.

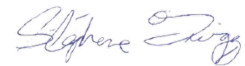
Water Stewardship Policy

Canadian Royalties Inc. is a mining company committed to protecting the environment and respecting communities. We recognize the importance of responsible water use and adhere to the principles promoted by the Mining Association of Canada's *Towards Sustainable Mining* initiative, including those of the Water Stewardship Framework. In pursuit of the commitments of its Social and Environmental Responsibility Policy, Canadian Royalties Inc. is committed to :

- Apply transparent corporate water governance, including:
 - ✓ Publicly disclose the company's approach to water stewardship
 - ✓ Allocate clear responsibilities and accountabilities for water
 - ✓ Integrate water considerations in business planning
 - ✓ Publicly report company water performance
- Manage water at operations effectively, including :
 - ✓ Maintain a water balance and understand how it relates to the cumulative impact of other users
 - ✓ Set context-relevant water targets or objectives for sites with material water-related risks
- Collaborate to achieve responsible and sustainable water use, including:
 - ✓ Identify, evaluate, and respond to catchment-level water-related risks and opportunities and engage with stakeholders that may influence or be affected by a site's water use and discharge
 - ✓ Support water stewardship initiatives that promote better water use, effective catchment management and contribute to improved water security and sanitation




Mathieu Roberge
General manager of Operations.
March 1st 2024


Stéphane Twigg
Environmental Director.
March 1st 2024